Public Health Analytics of AMR

EPSRC CDT in Engineering Solutions to Antimicrobial Resistance

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Outline

- Aims
- Data
- Health Economics
- Diagnostics
- Bayesian thinking
- Mechanistic modelling
- Conclusion

Resources

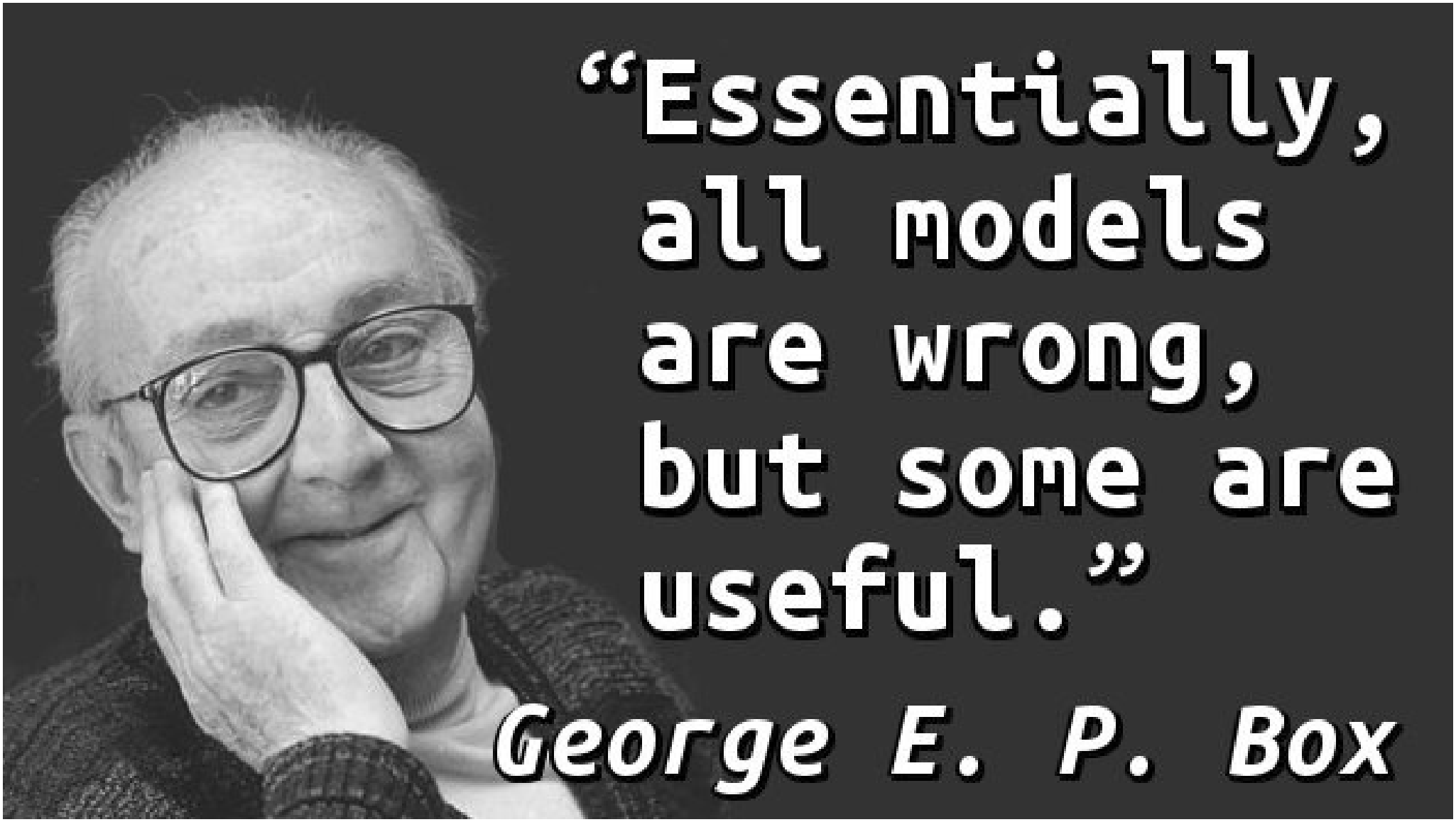
Slides and code here: `github.com/n8thangreen/AMR-presentation`

Aims

- How does data analysis, modelling, and public health policy intersect in tackling antimicrobial resistance.
- Provide insight into key concepts, challenges, and research directions
  - May encourage further interest and independent exploration.
- Develop a wide-ranging understanding of the AMR landscape
  - Preventing Infections
  - Diagnostics and Surveillance
  - Novel Therapies

Some Relevant Modelling

- Infectious disease modelling
- Survival analysis
- Probabilistic simulation
  - Cohort model (Markov model)
  - Discrete Event Simulation (DES)
  - Microsimulation etc
- Cost-effectiveness modelling, sensitivity analysis
- Missing data, linked data sets
- Causal inference


"Essentially, all models are wrong, but some are useful."
George E. P. Box

Data

Overview

What data should we look for?

- The role of **conceptual models**

Where will we find it?

- **Sources** and **forms** of data
- **Searching** for data
- **Selecting** data

What do we do with it?

- Evidence synthesis
- Data transformation

Steps for reviewing and using relevant data

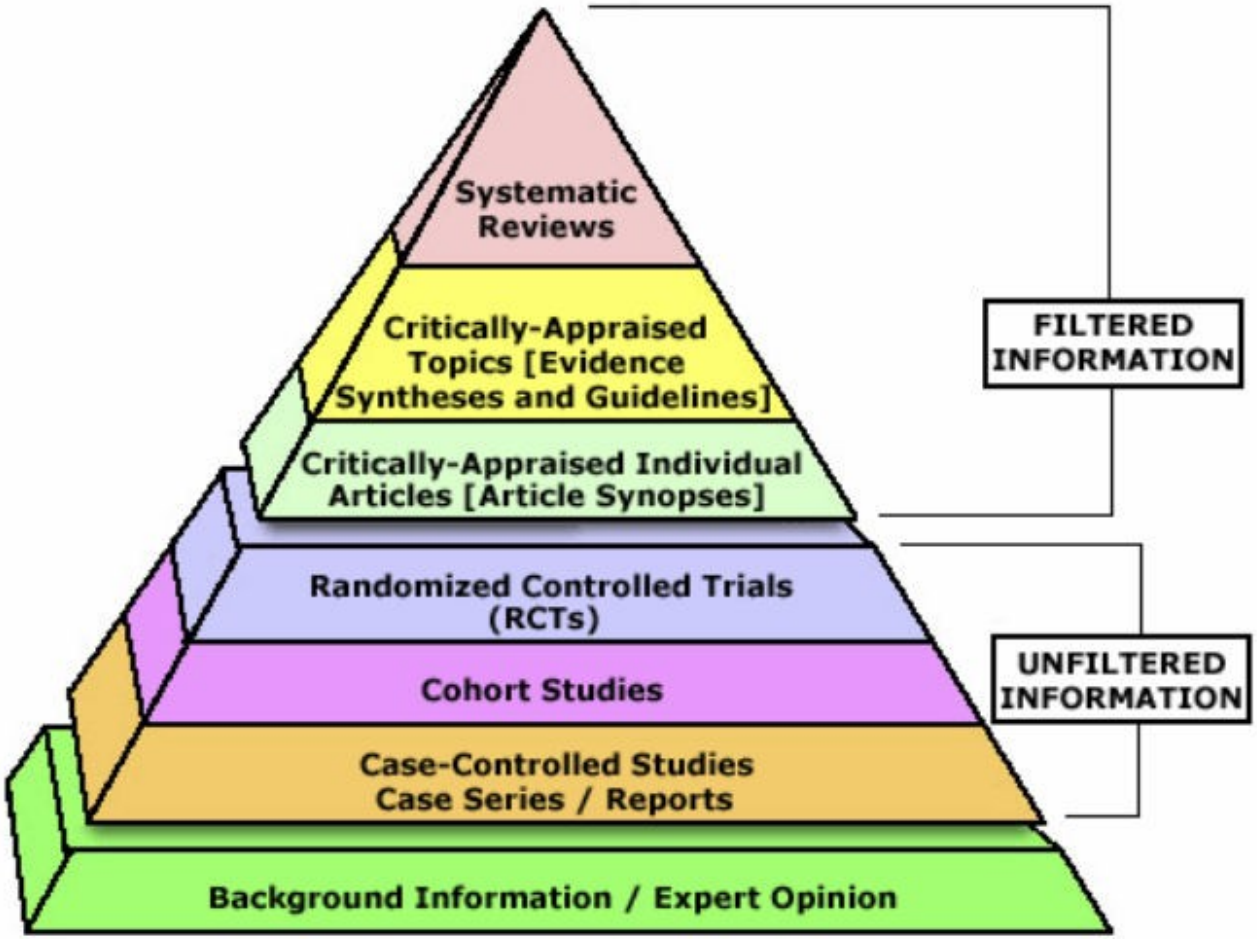
1. Consider how parameters relate to each other and group them
2. For each group, consider appropriate data sources
3. Critically assess the quality of the data sources
4. Take into consideration the different hierarchies of evidence

Critical assessment of data sources

Questions to consider.

1. Primary vs. secondary data?
2. Randomised vs. observational data?
3. Aggregate statistics vs. individual patient data?
4. When collected?
5. Sample size
6. Heterogeneity, case mix

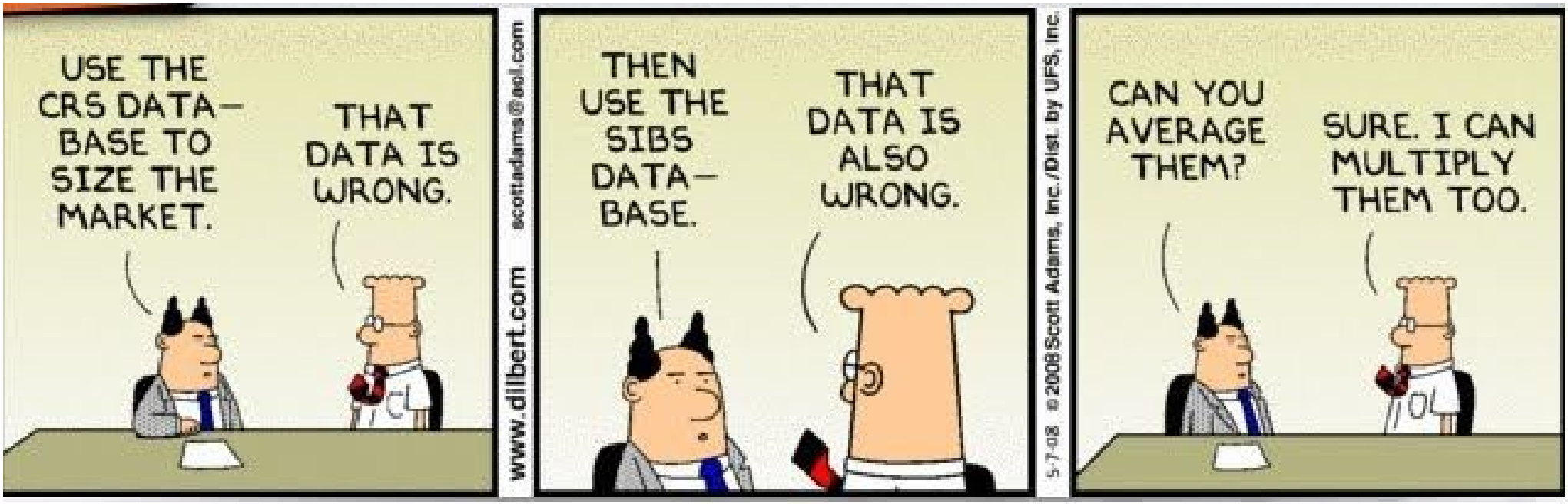
Hierarchy of evidence pyramid



The Global and Local Data for AMR

- **Global level:** WHO GLASS (Global Antimicrobial Resistance and Use Surveillance System).
- **National level:** UKHSA (UK Health Security Agency), ESPAUR reports.
- **Local/Clinical level:** Hospital Electronic Health Records (EHRs), Laboratory Information Management Systems (LIMS).
- **The Surveillance Dichotomy:**
  - *Passive Surveillance:* Waiting for routine clinical samples (cheap, scalable, but biased).
  - *Active Surveillance:* Deliberately swabbing populations or environments (comprehensive, but expensive and resource-intensive).

Messy, Noisy Data



Messy, Noisy Data

- **Selection Bias:** May mainly culture samples from patients who are severely ill or failing first-line therapy
- **Missingness & Fragmentation:**
  - Rubin's classification: Missing **Completely** At Random (MCAR), Missing At Random (MAR), Missing **Not** At Random (MAR)
- **Temporal Lags:** E.g. The delay between infection onset, laboratory culturing, and reporting
- **The Prediction Problem:** High dimensionality

Texas Sharpshooter Fallacy

Selection Bias

Operation Hummingbird | Exhibit Ref: CEH/16 | Staff Presence Report

Chart 1: Nurses present on clinical and administrative duties Summary

				Nursing Staff on Clinical and Administrative Duties																																			
Event	Baby	Date Shift Started	Shift Type	Alka SIMPSON		Angela McSHANE	Ashleigh HUDSON	Belinda SIMCOCK	Bernadette BUTTERWORTH	Caroline BENNION		Caroline OAKLEY	Cheryl CUTHBERTSON-TAYLOR	Christopher BOOTH	Eirian POWELL	Elizabeth MARSHALL		Janet COX	Jean PEERS	Jennifer JONES-KEY	Joanne WILLIAMS	Kathryn WARD		Laura EAGLES	Lisa WALKER	Lucy LETBY	Mary GRIFFITH	Melanie TAYLOR	Minna LAPPALAINEN	Nicola DENNISON	Patricia STEELE	Rebecca MORGAN	Samantha O'BRIEN	Shelley TOMLINS	Sophie ELLIS		Valerie PARKES	Valerie THOMAS	Vicky BLAMIRE	Yvonne FARMER	Yvonne GRIFFITHS
1	Child A	08/06/2015	NIGHT							X						X									X	X	X	X							X						
2	Child B	09/06/2015	NIGHT										X									X			X	X	X								X						
3	Child C	13/06/2015	NIGHT													X	X									X		X		X				X							
4	Child D	21/06/2015	NIGHT								X	X				X						X				X															
5	Child E	03/08/2015	NIGHT					X				X													X	X											X				
6	Child F	04/08/2015	NIGHT					X					X													X								X	X		X				
7	Child G	06/09/2015	NIGHT	X										X									X		X	X															
8	Child G	21/09/2015	DAY							X		X			X		X	X								X					X			X			X			X	X
9	Child H	25/09/2015	NIGHT													X										X									X		X				X
10	Child H	26/09/2015	NIGHT											X		X	X									X							X								
11	Child I	30/09/2015	DAY				X		X						X		X		X						X	X		X			X					X				X	
12	Child I	12/10/2015	NIGHT				X					X						X						X		X															
13	Child I	13/10/2015	NIGHT									X						X			X			X		X							X							X	
14	Child I	22/10/2015	NIGHT				X							X												X		X									X				
15	Child J	26/11/2015	NIGHT									X												X		X	X			X									X		
16	Child J	17/12/2015	DAY												X				X		X			X		X			X		X								X		X
17	Child K	16/02/2016	NIGHT									X									X					X								X			X				
18	Child M	09/04/2016	DAY			X	X	X									X									X	X														
19	Child L	09/04/2016	DAY			X	X	X									X									X	X				X										
20	Child N	02/06/2016	NIGHT											X												X		X						X			X				
21	Child N	14/06/2016	NIGHT					X	X											X		X	X		X	X															
22	Child O	23/06/2016	DAY								X			X				X								X		X			X	X	X				X			X	X
23	Child P	23/06/2016	DAY								X			X				X								X		X			X	X	X				X			X	X
24	Child P	24/06/2016	DAY		X									X				X								X	X					X					X			X	
25	Child Q	25/06/2016	DAY														X									X	X		X	X											
			Total	1	1	2	5	5	2	2	3	7	2	7	3	5	7	6	2	1	3	3	2	4	6	25	7	7	2	3	6	3	2	3	4	4	5	6	2	6	5

*('X' indicates 'on duty' presence on the shift, where a suspicious event has been identified)

Predicting is hard

- *“Those who have knowledge, don’t predict. Those who predict, don’t have knowledge.”* — Lao Tzu (Philosopher)
- *“People can come up with statistics to prove anything, Kent. Forty percent of all people know that.”* — Homer Simpson

07:58 Samstag 21. Dez. 27 %

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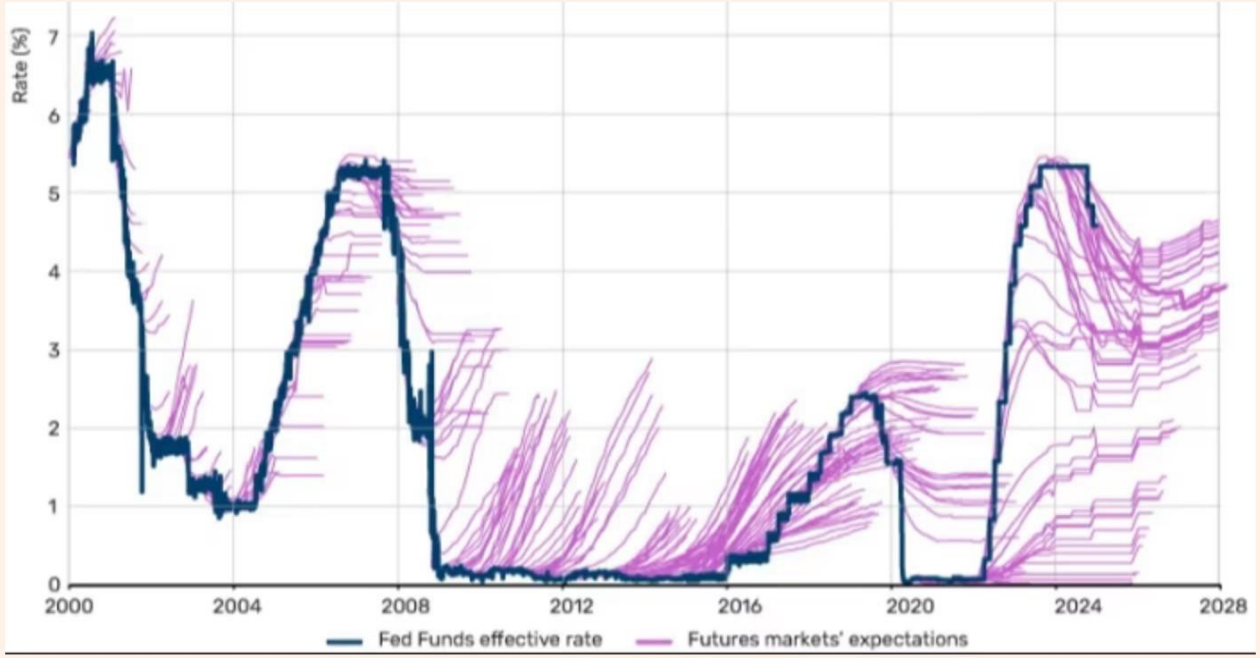
However, we mainly wanted to highlight the report for a beautiful, updated version of one of Alphaville's favourite-ever chart formats: implied rates forecasts over time, and what rates have actually done.



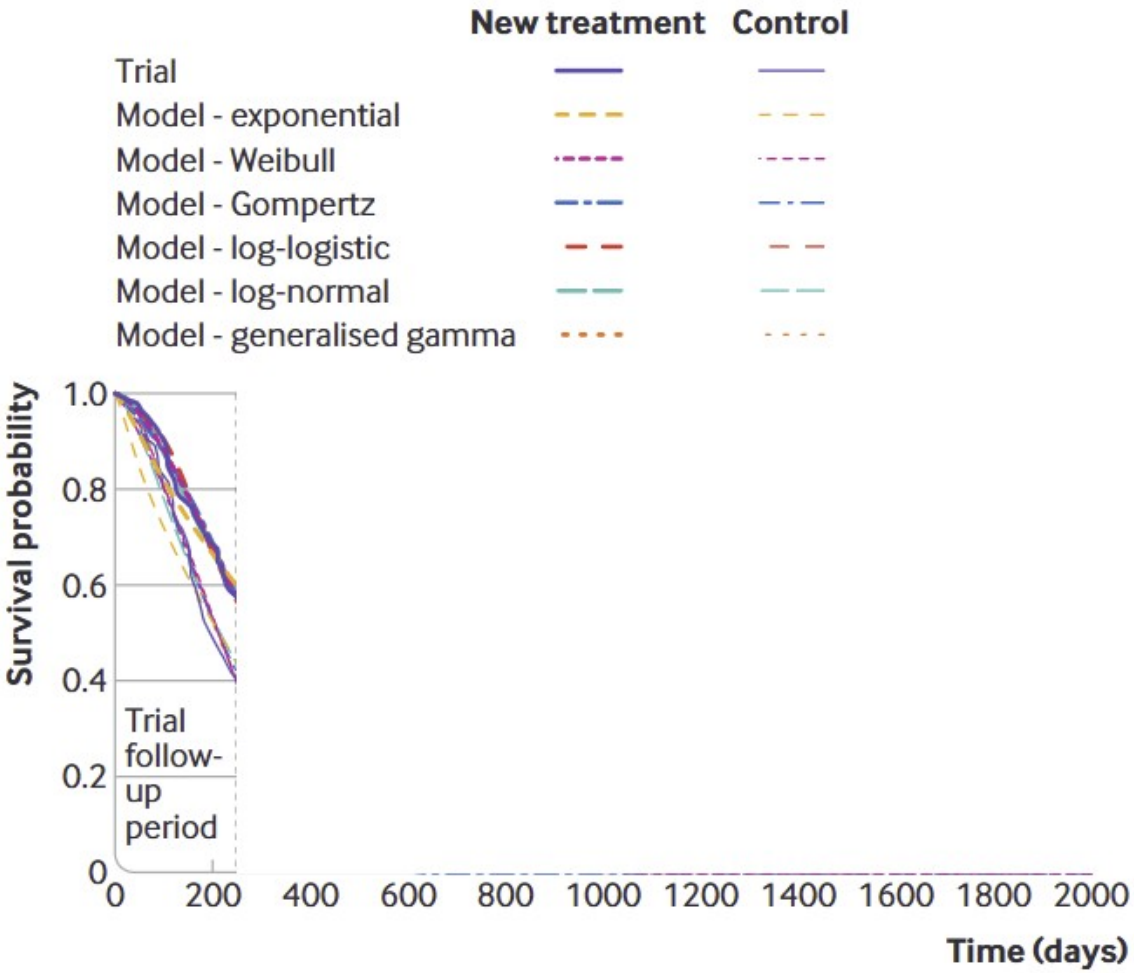
It's an oldie but a goody, and we reckon it's good to resurface from time to time as a reminder of how hard it is to make predictions, especially about the future. Please bear that in mind as we enter 2025!

What can we do about it?

- Multiple imputation
- Bayesian modelling
- Combining multiple data sets
  - Mutiparameter evidence synthesis (MPES)
  - (Post) Stratification
  - (Network) meta-analysis (NMA)
  - Extrapolation
  - …

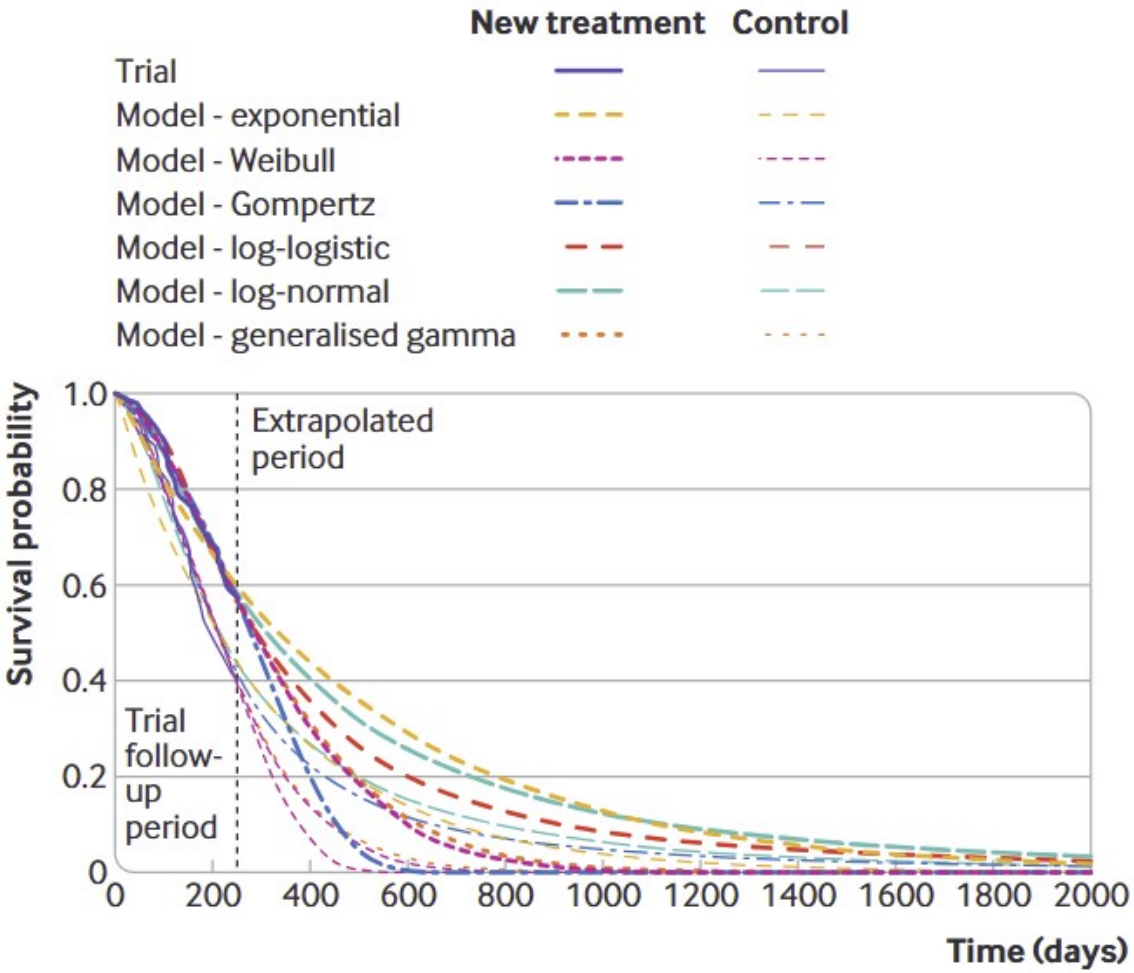
Survival curve extrapolation

Latimer and Adler (2022)



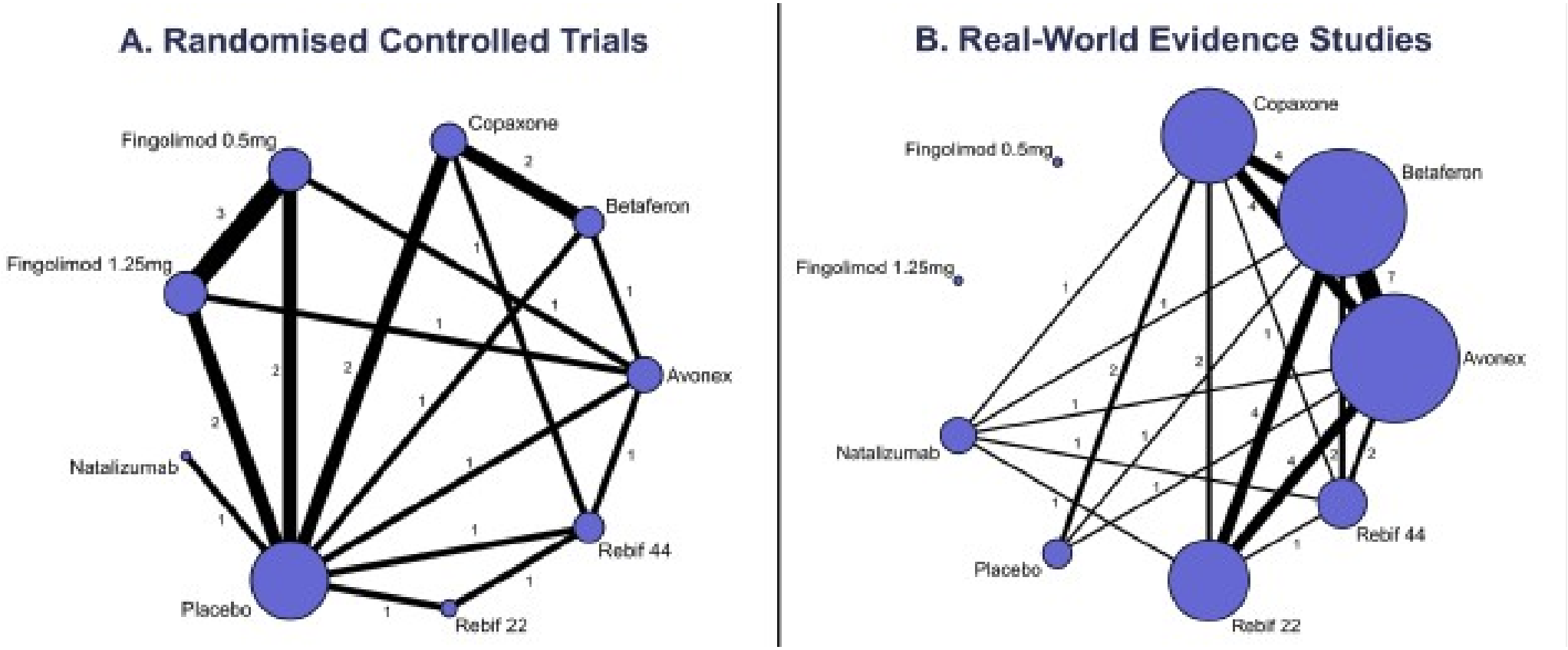
Survival curve extrapolation

Latimer and Adler (2022)



NMA

Jenkins et al. (2021)



Health Economics of AMR

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Health Economic Evaluation (HEE) / Health Technology Assessment (HTA)

- **Goal:** Maximize population health under a strictly constrained budget.
- **ICER (Incremental Cost-Effectiveness Ratio):**
  - $\mathrm{ICER} = \frac{\mathrm{Cost}_{\mathrm{new}} - \mathrm{Cost}_{\mathrm{standard}}}{\mathrm{Effect}_{\mathrm{new}} - \mathrm{Effect}_{\mathrm{standard}}}$
- **Cost-Effectiveness Plane:** Plotting $\Delta\mathrm{Cost}$ against $\Delta\mathrm{Effect}$
- **Willingness to Pay Threshold:** How bodies like NICE (National Institute for Health and Care Excellence) make decisions. Historically, the NHS threshold is £20,000 – £30,000 per QALY gained.

Recent change in theshold

NICE National Institute for Health and Care Excellence

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Changes to NICE's cost-effectiveness thresholds confirmed

The government has today (1 December 2025) announced that it will increase the thresholds NICE uses in evaluations of new medicines to decide which are cost-effective for use in the NHS.

News 1 December 2025

Recent change in theshold

Pharmaceuticals industry

This article is more than **4 months old**

NHS could pay 25% more for medicines under plan to end row with drugmakers and Trump

UK ministers preparing to raise payments to end standoff with pharma industry, cited as reason for firms axing British investments

Julia Kollewe *and* Eleni Courea

Wed 8 Oct 2025 21.10 BST

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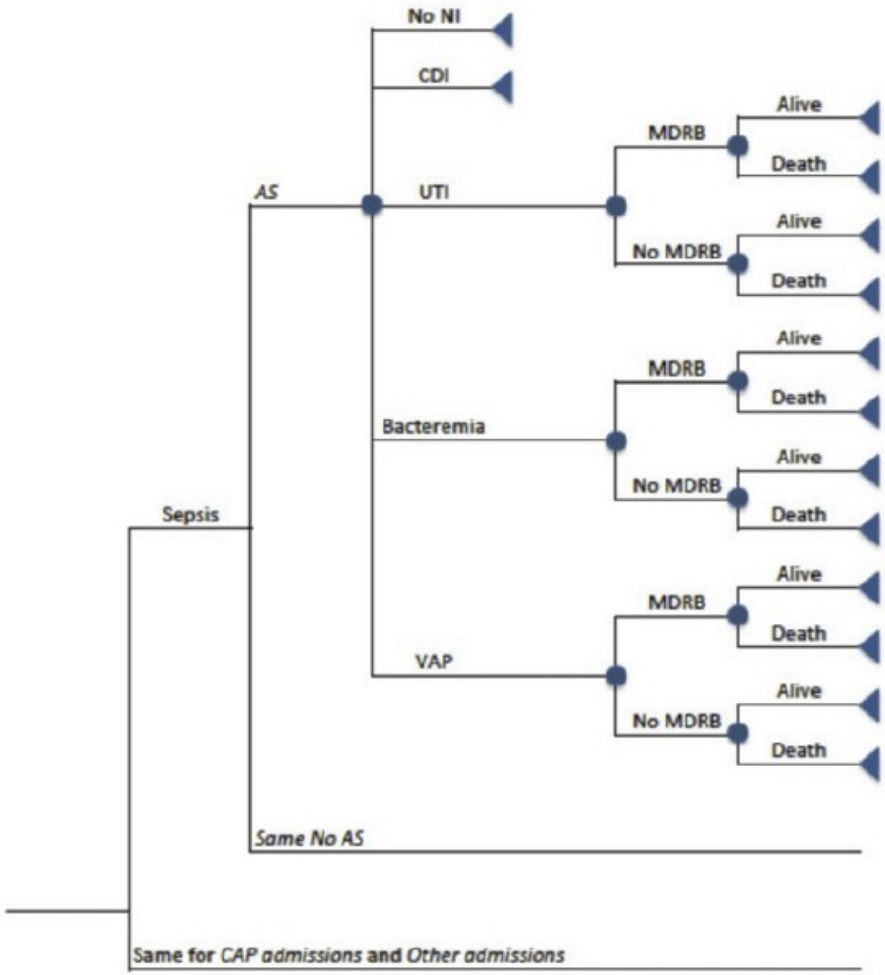
Measuring the Burden

- **QALYs (Quality-Adjusted Life Years):** Measures health gain ($1\ \mathrm{QALY} = 1\ \mathrm{year\ in\ perfect\ health}$).
- **DALYs (Disability-Adjusted Life Years):** Measures health loss (Years of Life Lost + Years Lived with Disability).
- **Direct Healthcare Costs:** Prolonged hospital stays, intensive care bed days, expensive last-resort antibiotics.
- **Indirect Societal Costs:** Lost productivity, macroeconomic impacts, and the “insurance value” of antibiotics (enabling surgeries, chemotherapy).

Ruiz-Ramos et al. (2017)



AS: Antimicrobial stewardship; NI: Nosocomial infection; CDI: *Clostridium difficile* infection; UTI: Urinary tract infection; VAP: Ventilator-associated pneumonia; CAP: Community adquired pnemonia; MDRB: Multidrug-resistant bacteria

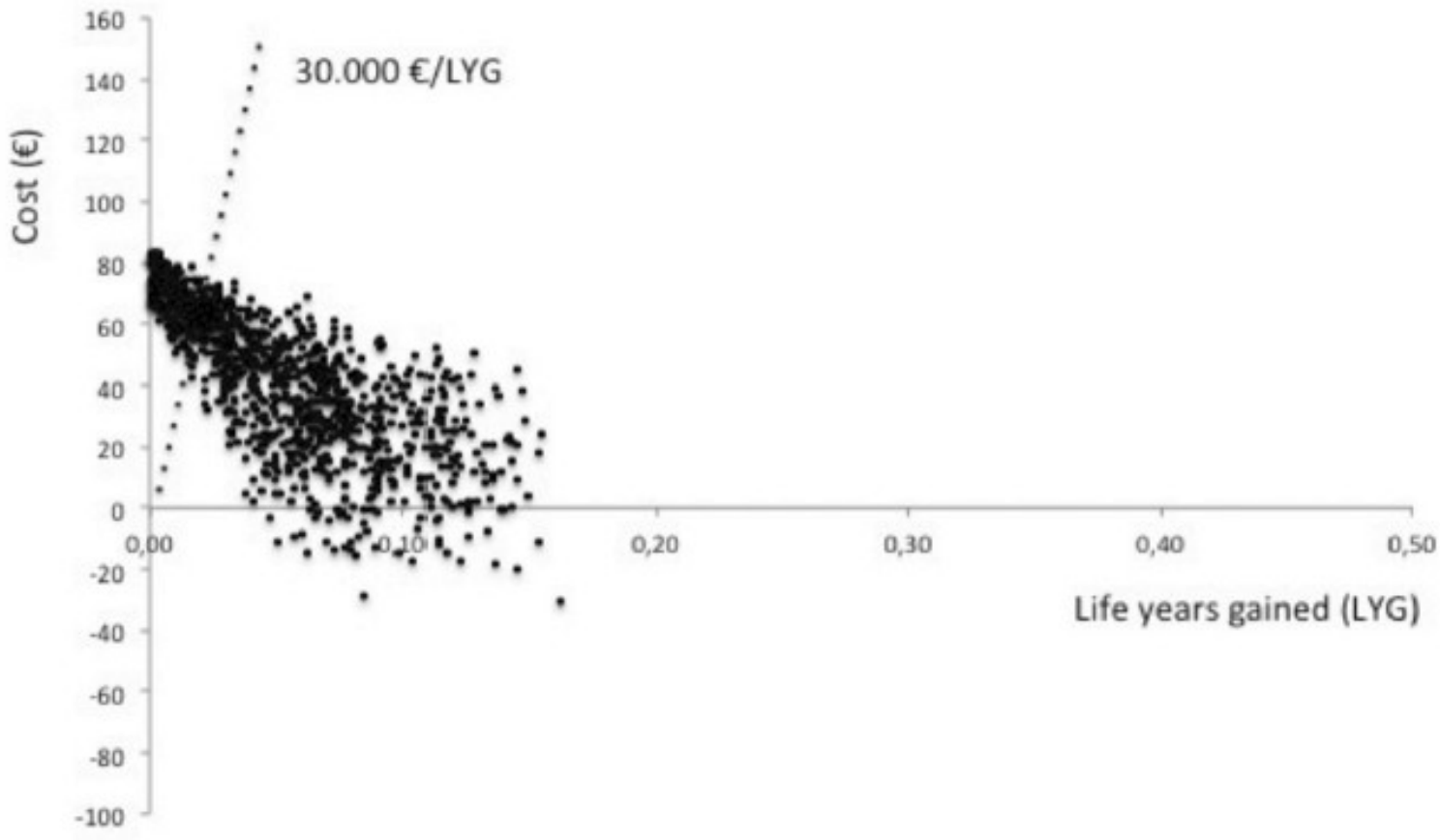
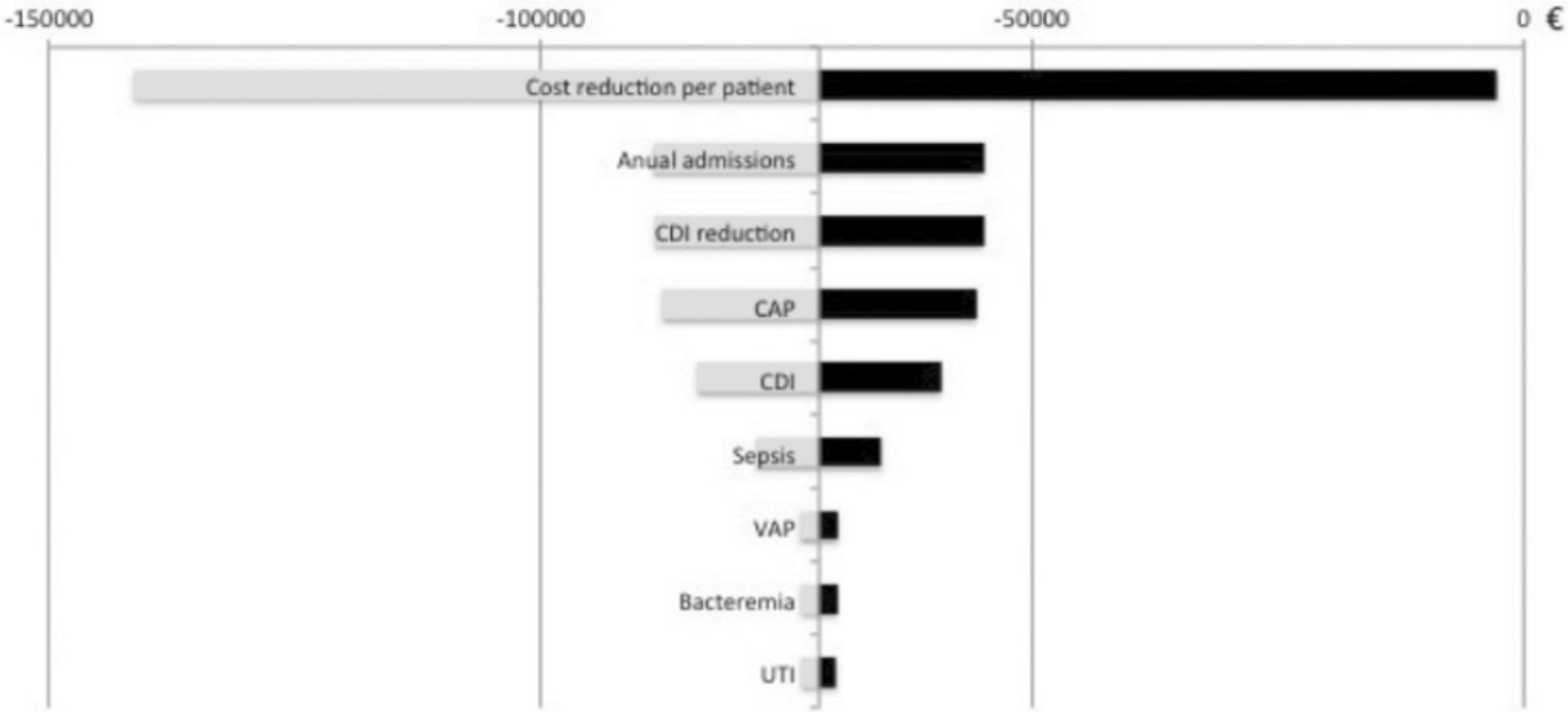


Figure 3. Long-term multivariate analysis results (Cost per LYG).

Deterministic Sensitivity Analysis (DSA)



CDI=*Clostridium difficile* infection; UTI: Urinary tract infection; VAP: Ventilated-associated pneumonia; CAP: Community adquired pneumonia.

Probablistic Sensitivity Analysis (PSA)

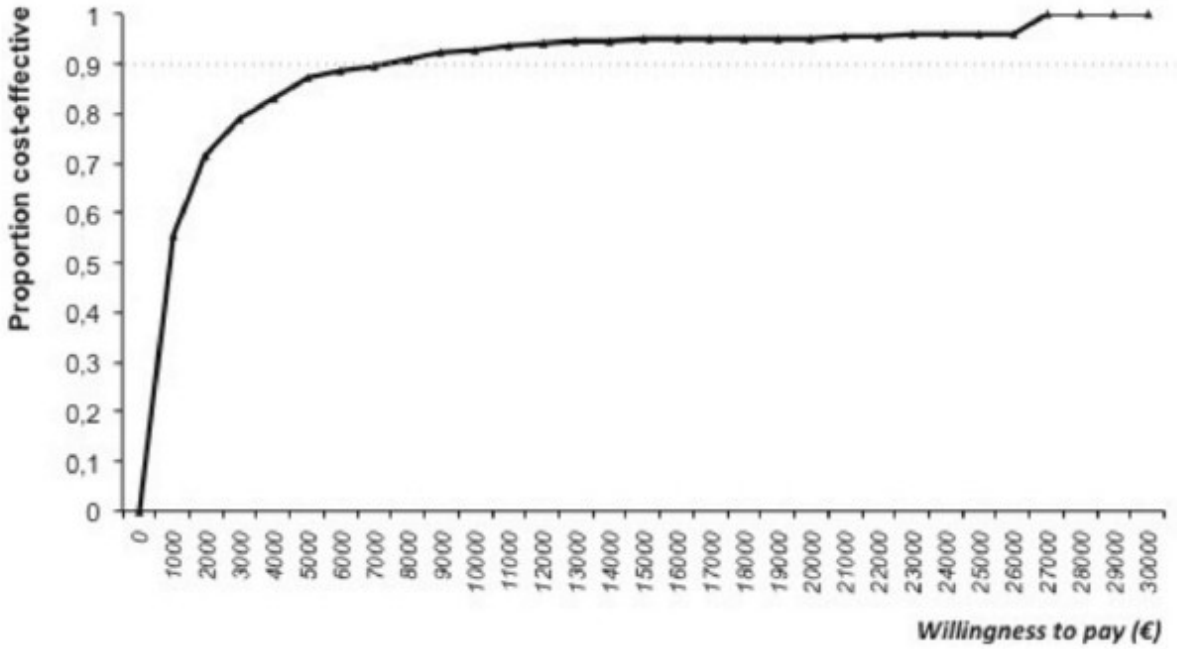


Figure 5. Predicted proportion of cost-effective antimicrobial stewardship (AS) services at incremental willingness to pay per LYG. Incremental cost-effectiveness of implementing an AS was calculated and compared with standard to compute an ICER.

BREAK 😊

Diagnostics

Statistical Analysis of Diagnostics

- **Clinical Reality:** A rapid test is only as good as the clinical action it prompts.
- **Inherent Properties:** Sensitivity (true positive rate) and Specificity (true negative rate).
- **Real-World Performance:** Positive/Negative Predictive Values (PPV/NPV).
- PPV is highly dependent on *background prevalence*.
  - Highly specific device can still yield mostly false positives if deployed in a low-prevalence population!

Fundamental classification statistics

- **True positive (TP)**: Number of correct predicted positives.
- **True negative (TN)**: Number of correct predicted negatived.
- **False positive (FP)**: Number of incorrect predicted positives. Equivalent with Type I error.
- **False negative (FN)**: Number of incorrect predicted negatives. Equivalent with Type II error.

Contingency tables

	True Positive (p)	**True Negative (n)**	**Total**
Test Positive (p’)	True Positive (TP)	False Positive (FP)	**P’**
Test Negative (n’)	False Negative (FN)	True Negative (TN)	**N’**
Total	**P**	**N**	

Contingency table summary statistics

- **Sensitivity** $= p(T|D) = \frac{TP}{TP+FN} = \frac{TP}{P}$
- **Specificity** $= p(T^c|D^c) = \frac{TN}{FP+TN} = \frac{TN}{N}$
- **Prevalence** $= \frac{TP+FN}{M} = \frac{P}{M}$
- **Positive predictive value (PPV)/precision** $= p(D|T) = \frac{TP}{TP+FP}$
- **False discovery rate (FDR)** $= 1 - PPV = p(D^c|T) = \frac{FP}{TP+FP}$
- **Negative predictive value (NPV)** $= p(D^c|T^c) = \frac{TN}{FN+TN}$
- **False omission rate (FOR)** = 1-NPV $= p(D|T^c) = \frac{FN}{FN+TN}$
- **Accuracy** $= \frac{TP+TN}{M}$

Contingency table summary statistics

Notice that the PPV and NPV depend on the prevalence. We can alternatively compute them using:

- **PPV** =

$$\frac{sensitivity \times prevalence}{sensitivity \times prevalence + (1 - specificity) \times (1 - prevalence)}$$

- **NPV** =

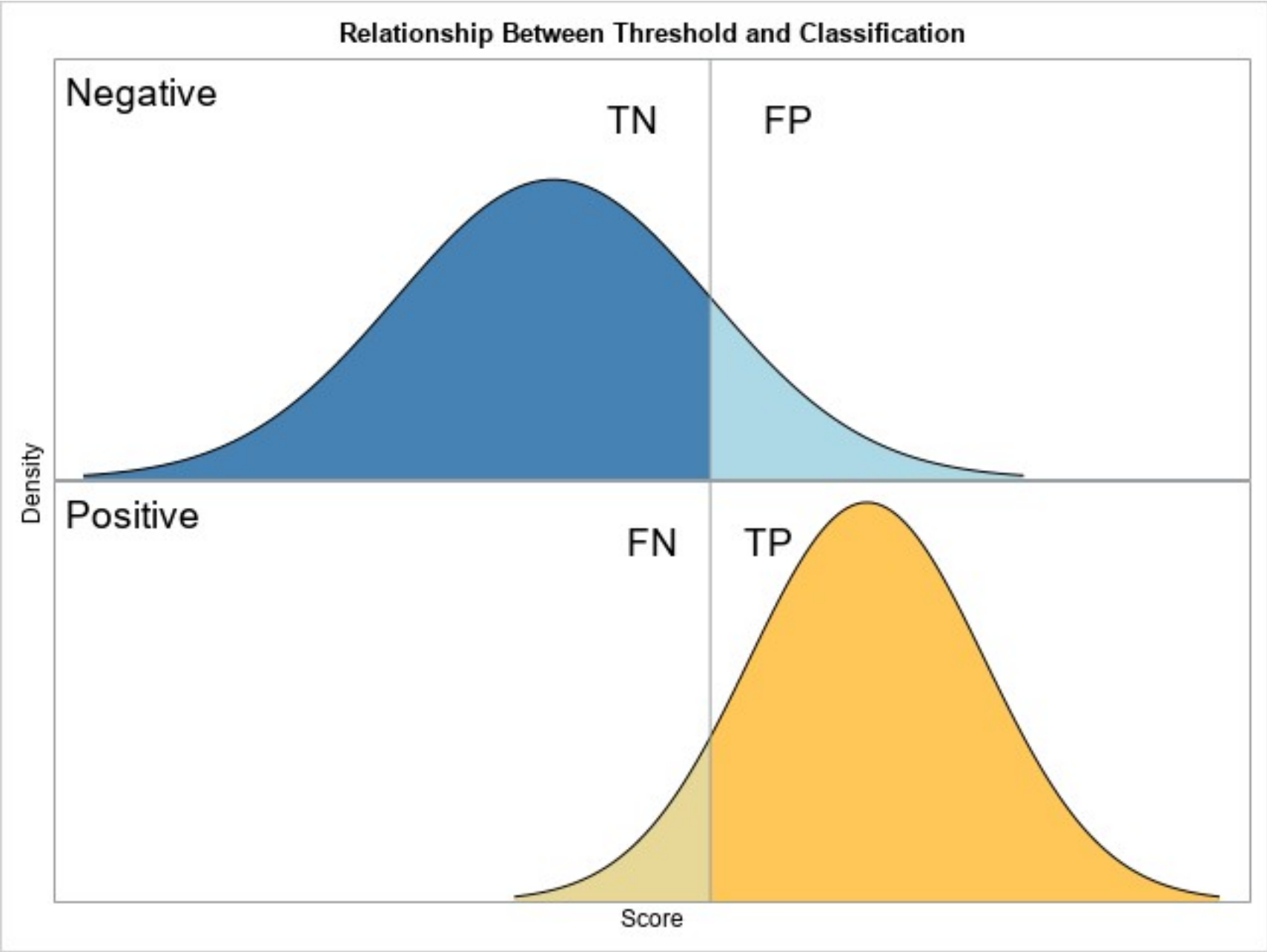
$$\frac{specificity \times (1 - prevalence)}{(1 - sensitivity) \times prevalence + specificity \times (1 - prevalence)}$$

How to compare the performance of diagnostic tests?

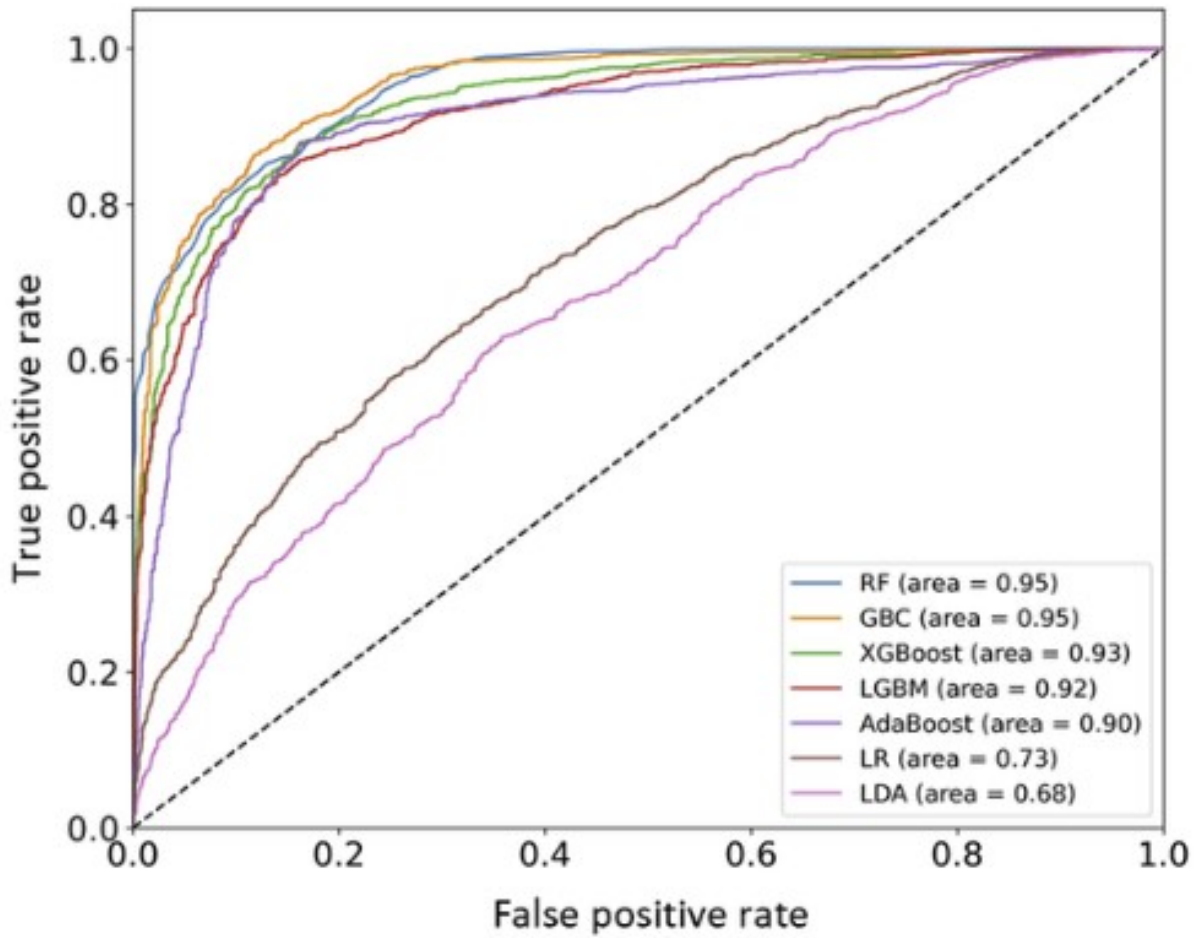
- Dichotomous test (only 2 results)
- Odds ratios
- Likelihood ratios
- Sensitivity specificity, PPV, NPV
- Multilevel level (> 2 results)
- Receiver operating characteristic curve (ROC)

Receiver operating characteristic curves (ROC)

- TP rate (sensitivity) on vertical axis and FP rate (1-specificity) on horizontal, for varying classification threshold.
- Illustrates the trade-off between sensitivity and specificity.
- A single curve summary of the information in the cumulative distribution functions of the scores of the two classes.
- c.f. *ROC Curves for Continuous Data*, Krzanowski and Hand (2009).


Relationship Between Threshold and Classification
Negative
TN
FP
Positive
FN
TP
Density
Score

Krzanowski and Hand (2009)



ROC curves for predicting antibiotic resistance in Klebsiella pneumoniae. (A) ROC curves for CIP resistance prediction. (B) ROC curves for LEV resistance prediction. ROC: receiver operating characteristic; CIP: ciprofloxacin; LEV: levofloxacin; RF: random forest; GBC: gradient boosting classifier; LGBM: light gradient boosting machine; LR: logistic regression; LDA: linear discriminant analysis.

Bayesian Thinking

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Frequentist vs. Bayesian:*

- *Frequentist:* The true rate of AMR is fixed; our data is a random sample.
- *Bayesian:* Our *belief* about the rate of AMR is a probability distribution, which we update as we get new data.
- Start with a prior belief, observe new evidence (e.g., from your engineered biosensor), and update to a posterior belief.
- Diagnostic devices do not give absolute answers; they provide *evidence* that shifts a clinical probability.

Bayes' Theorem for Diagnostics

- **The Equation:**

$$P(\text{AMR}|\text{Positive Test}) = \frac{P(\text{Positive Test}|\text{AMR}) \cdot P(\text{AMR})}{P(\text{Positive Test})}$$

- **Breaking it down:**
  - $P(\text{AMR}|\text{Positive Test})$: **Posterior** (What the clinician actually wants to know—the PPV).
  - $P(\text{Positive Test}|\text{AMR})$: **Likelihood** (Your device's Sensitivity).
  - $P(\text{AMR})$: **Prior** (Background prevalence of this resistant bug in this specific hospital).
  - $P(\text{Positive Test})$: **Marginal Likelihood** (Probability of the device firing positive at all).

Denominator Bias

- **False Positive Paradox:** Let's say you engineer a biosensor that is 99% sensitive and 99% specific.
- You deploy it in a general ward where the prior probability (prevalence) of a specific AMR strain is only 1%.
- **Reality Check:** If the alarm rings, what is the probability the patient actually has the resistant bug?
  - It's only 50%. Half of your alarms will be false positives.
- **Takeaway:** Engineering a highly accurate device is not enough. If you deploy it in the wrong clinical context (the wrong prior), e.g. over-prescription of antibiotics, and ultimately fail HTA evaluation.

Mechanistic Modelling

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- **Challenge:** We cannot wait 10 years for a clinical trial to prove a new technology reduces AMR at the population level.
  - Indirect interactions
  - Mass action
- **Mechanistic Models (Transmission Dynamics):** - Adapting SIR (Susceptible-Infected-Recovered) models to hospital wards.
  - E.g. simulating how a rapid diagnostic cuts the chain of transmission.

Susceptible
Infection
Infectious
Recovery
Recovered

Durazzi et al. (2023)

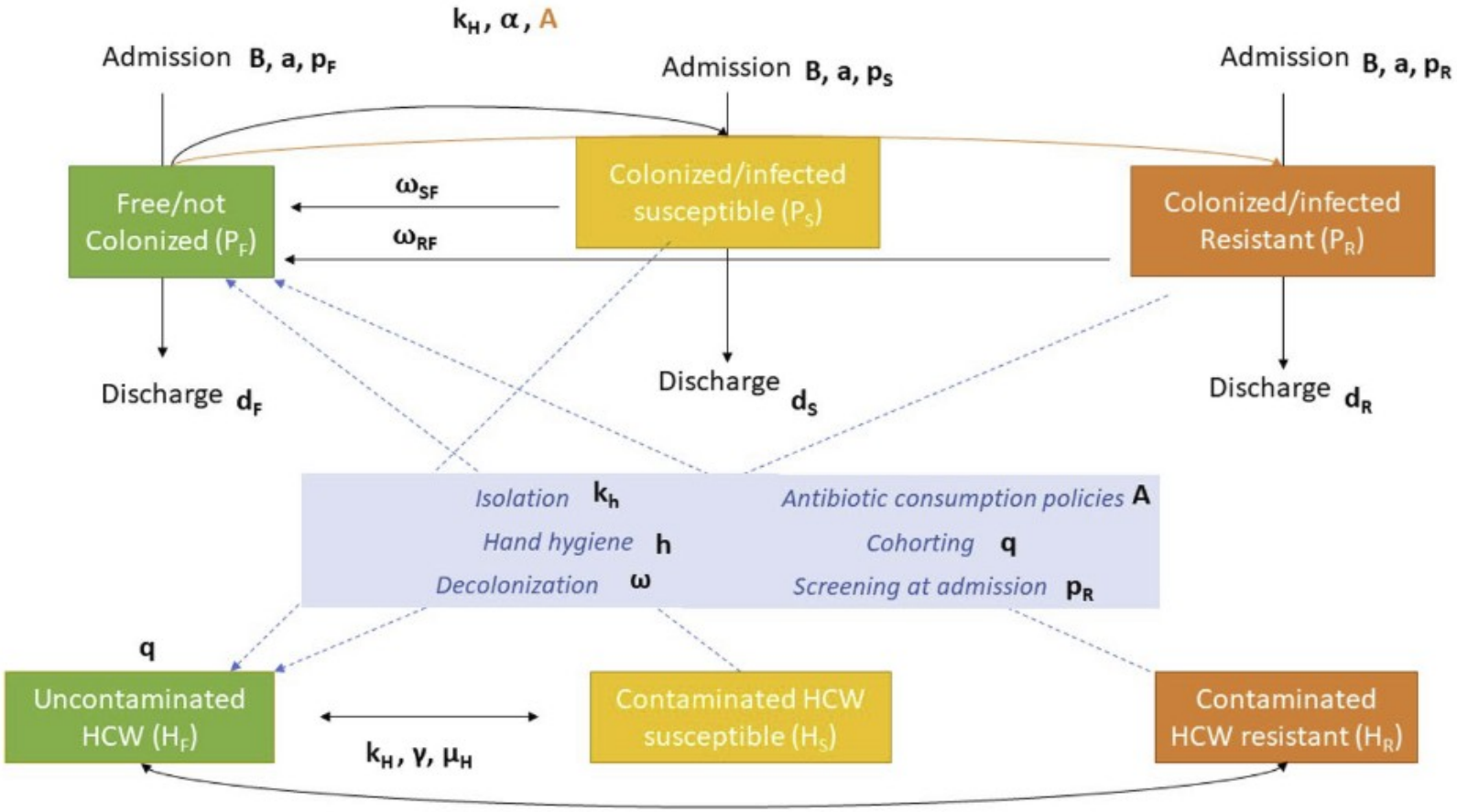
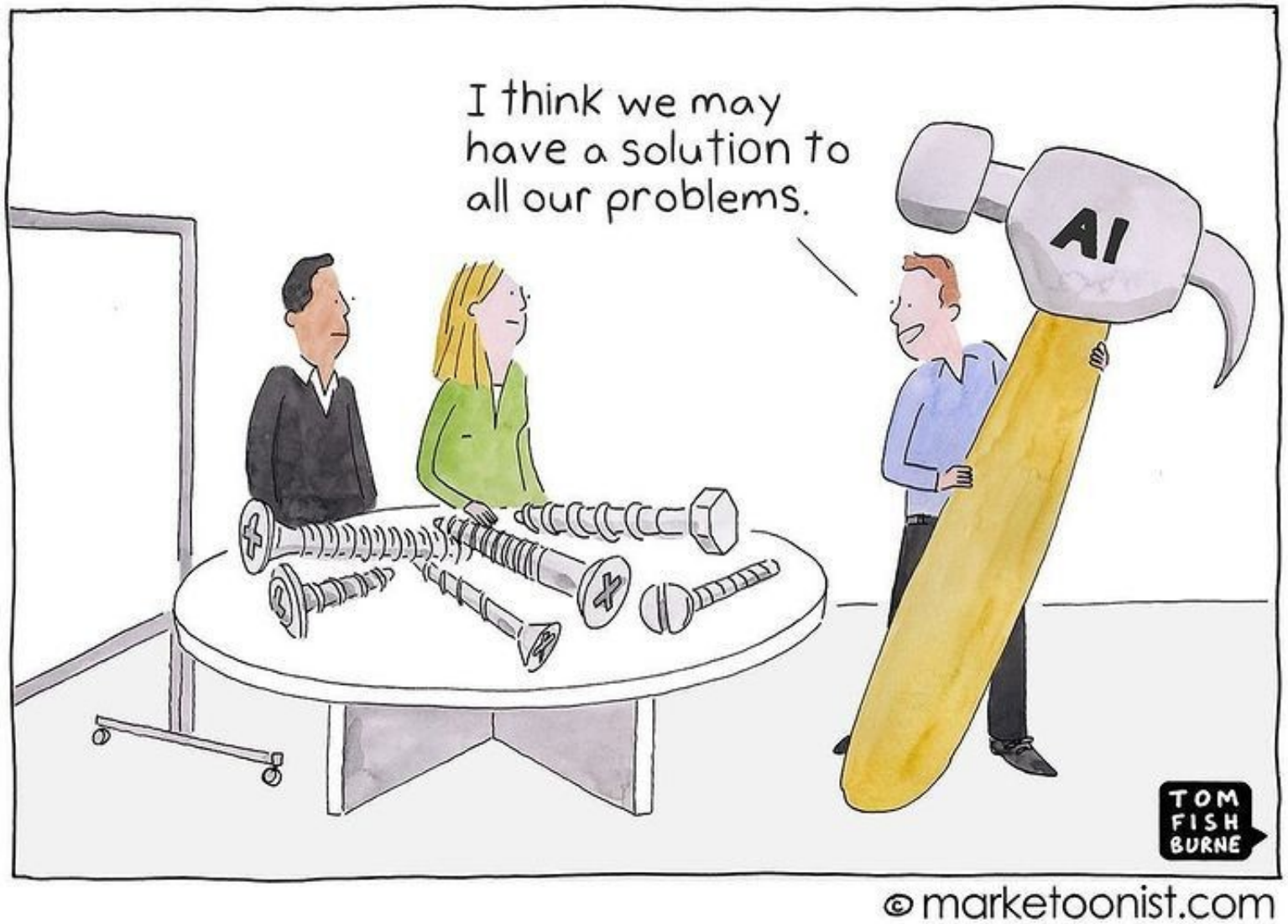


Figure 1. Transmission model: the epidemiological state of a patient belongs to one of three different categories: free/not colonized from the target pathogen (P_F), colonized/infected by susceptible strain (P_S), colonized/infected by resistant strains (P_R). Similarly, healthcare workers (HCWs) can be uncontaminated or contaminated with susceptible or resistant strains (HS and HR respectively).

System of differential equations

$$\delta P_F = -K_H \alpha A H_R P_F - K_H \alpha H_S P_F - d_{mean} P_F + a P_F + \omega_{RF} P_R + \omega_{SF} P_S$$
$$\delta P_S = K_H \alpha H_S P_F - d_{mean} P_S + a P_S - \omega_{SF} P_S$$
$$\delta P_R = K_H \alpha A H_R P_F - d_{mean} P_R + a P_R - \omega_{RF} P_R$$
$$\delta H_F = -K_H \gamma (1-h) H_F (P_S + P_R) + K_H h (H_S + H_R)(P_F + P_S + P_R) + \mu_H (H_S + H_R)$$
$$\delta H_S = K_H \gamma (1-h) H_F P_S - K_H h H_S (P_F + P_S + P_R) - \mu_H H_S$$
$$\delta H_R = K_H \gamma (1-h) H_F P_R - K_H h H_R (P_F + P_S + P_R) - \mu_H H_R$$

A comment on AI



Mentimeter

brief quiz (3–5 multiple-choice or short-answer questions)

Conclusions

- Models can describe, explain and predict
  - Machine learning are more black box and designed for prediction
- **Value is multifaceted:** Cost-effectiveness, not just clinical efficacy
- **Engage early:** Engineers, statisticians, and health economists must collaborate during the design phase of a technology, not just at the end.
- *"To consult the statistician after an experiment is finished is often merely to ask him to conduct a post mortem examination. He can perhaps say what the experiment died of."* — Ronald A. Fisher (Presidential Address to the First Indian Statistical Congress, 1938)

Thanks 🙏

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